⇒ Logistic Regression

⇓

'Classification Problem'

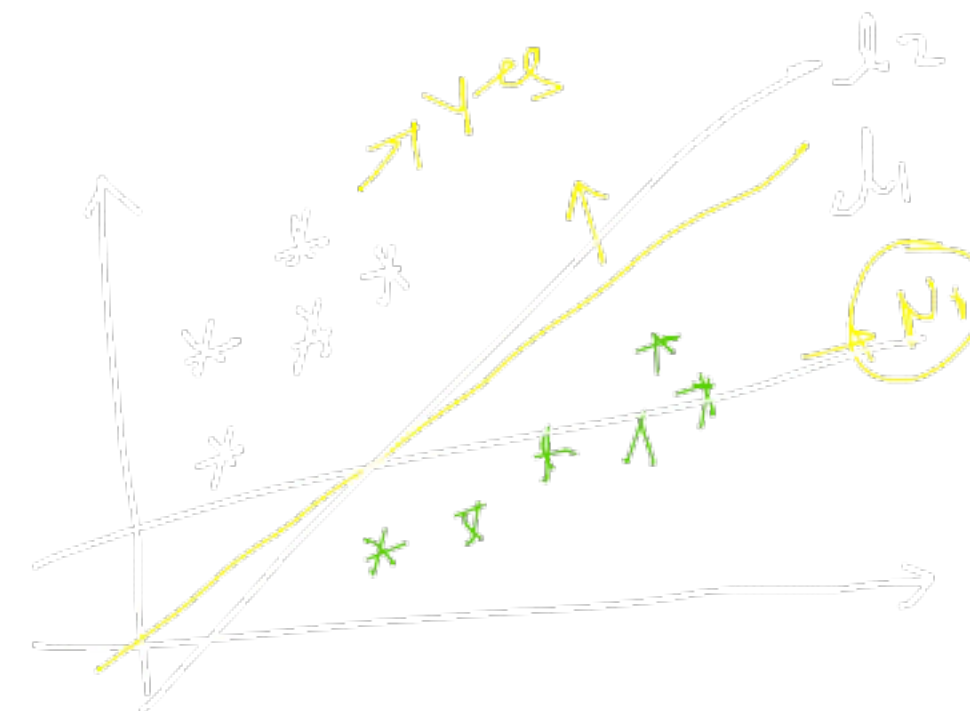
(i) Geometry

(ii) Probability

(iii) log-loss

EDA — ins

(case i)



data

Assumption → (1) linearly separable

(case ii)

2D

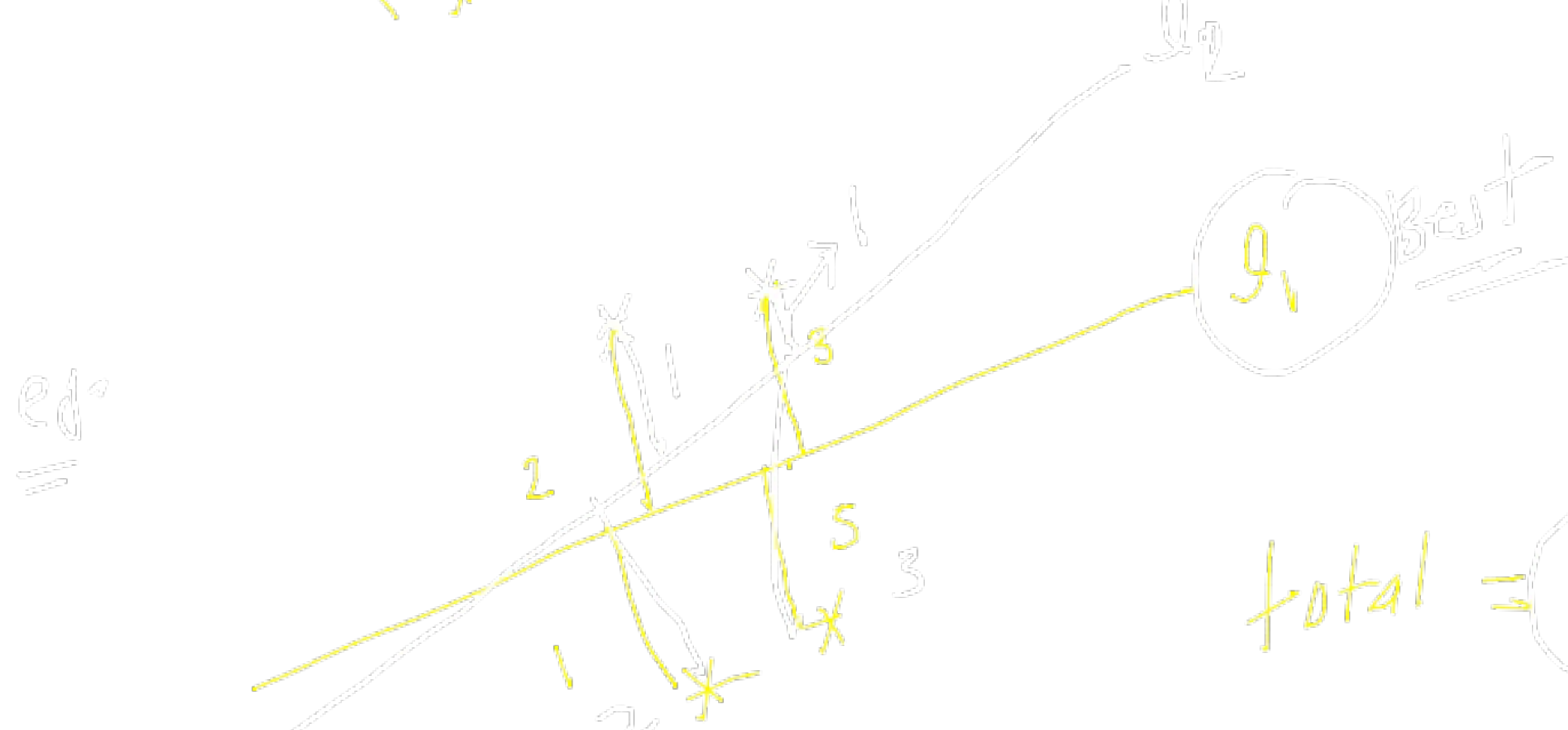
↓

line

LR

↓

Objective → Best fit line / plane → which can maximize the distance from plane / line



total = 11

total => 8

Equation of plane

$y = mx + 1$

$$\pi = w^T \cdot x + b$$

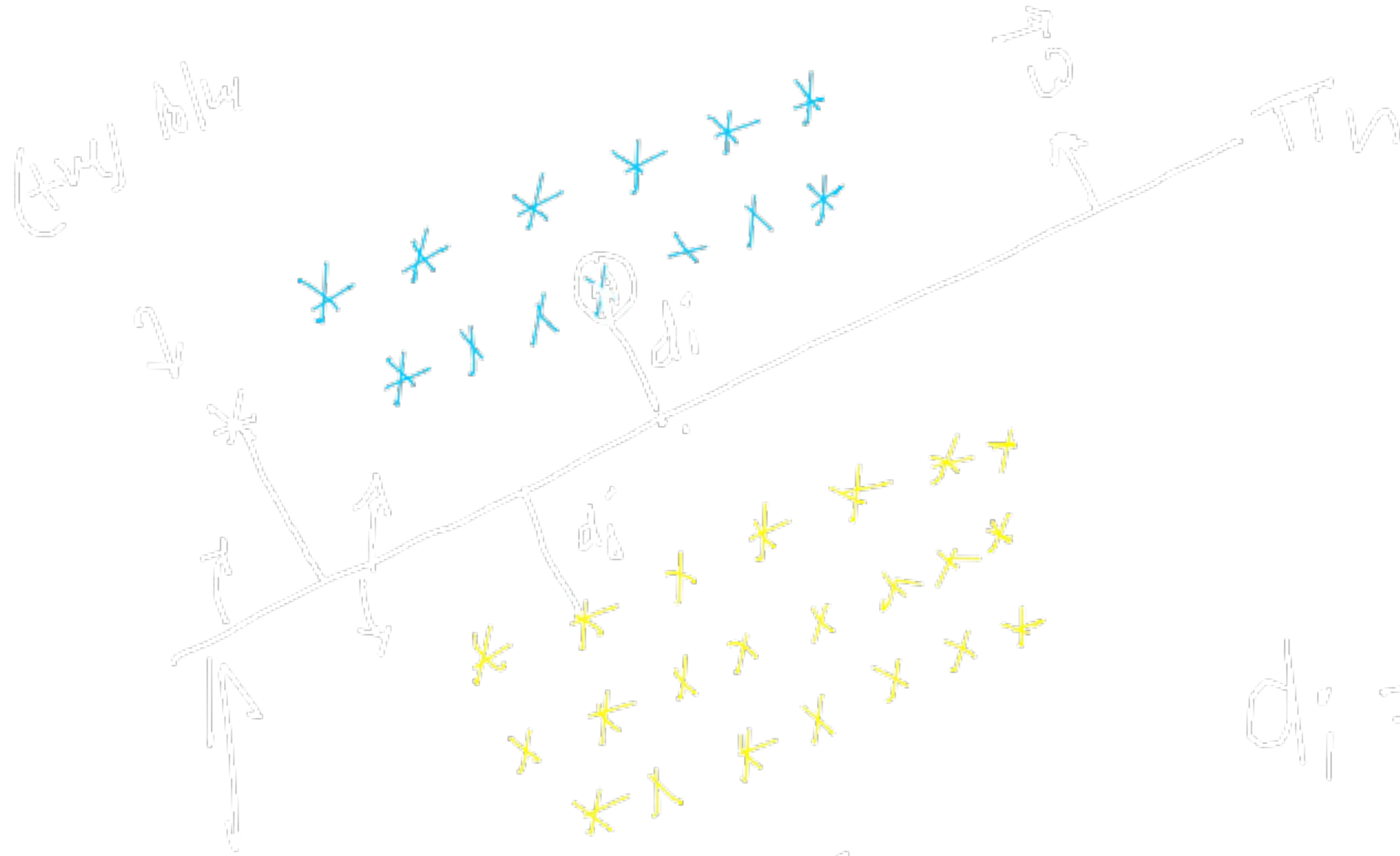
[0, 0.1, 0.2] [- -]

$D_n \Rightarrow \{x_i\ y_i\}$

Task find out $\boxed{w \ \& \ b}$

Yes No

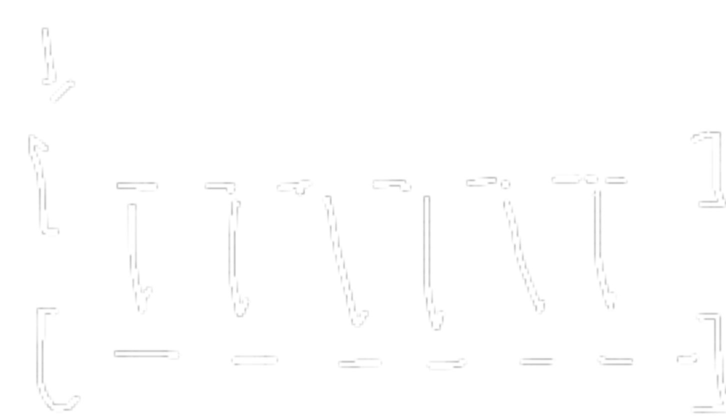
such that it separate +ve pts & -ve pts with maximum distance

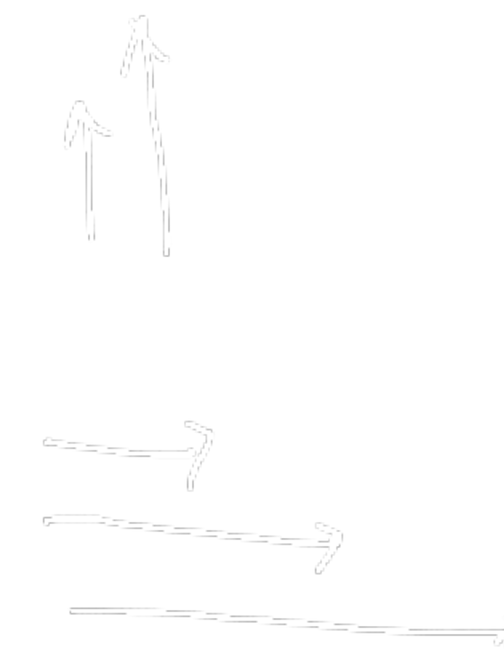


w. vector

w

x_i





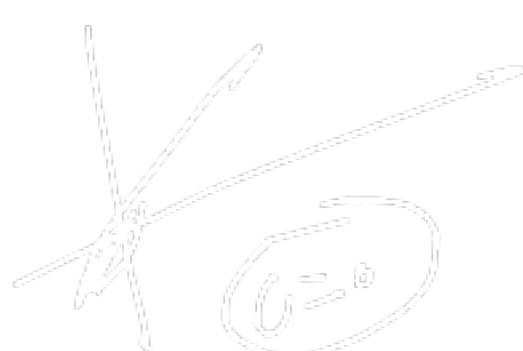
$$d_i = \frac{w^T \cdot x_i}{\|w\|}$$

→ identity ~~unit~~ vector → 1

$$\hat{w} = \frac{w}{|w|}$$

'Decision' surface

if $\boxed{\|w\| = 1}$



$$\rightarrow \begin{cases} d_i = w^T \cdot x_i \rightarrow +ve \\ d_j = w^T x_j \rightarrow -ve \end{cases}$$

Classifier say → Decision surface

if $y_i \cdot w^T \cdot x_i > 0$

$y_i \to$ (+ve) yes

if $w^T \cdot x_i < 0$

$\to y_i = (-ve)$

$\underset{\text{Actual}}{y_i} * w^T \cdot x_i > 0$

if $y_i * w^T x_i > 0$ → correct classifications

$(+ve) * (+ve) > 0$

→ correct classification

$(-ve)(-ve)$

↓

$(-ve)$

$y_i\, w^T \cdot x_i$

$(-ve)\,(+ve)$ → wrong

$(+ve)\,(-0.85)$

$(-ve)$

↓ $(-ve)$

Skewed

$y_i \cdot w^T x_i$

$(-ve)\,(-ve)$

+ve

y_1	+ve
y_2	+ve
y_3	-ve
y_4	-ve

$d = w^T \cdot x_i$

signed distance

max $\quad \underset{w}{\arg\max}\ d \uparrow$

$\Downarrow$ math

optimization problem

$$\underset{w^*}{\arg\max} \left(\sum_{i=1}^{n} y_i \cdot \underline{w^T \cdot x_i} \right) \quad \text{--- } (1)$$

Data

such that $y_i w^T \cdot x_i > 0 \quad \forall x_i$

$w^* = \arg\max \sum_{i=1}^{n} y_i w^T x_i$

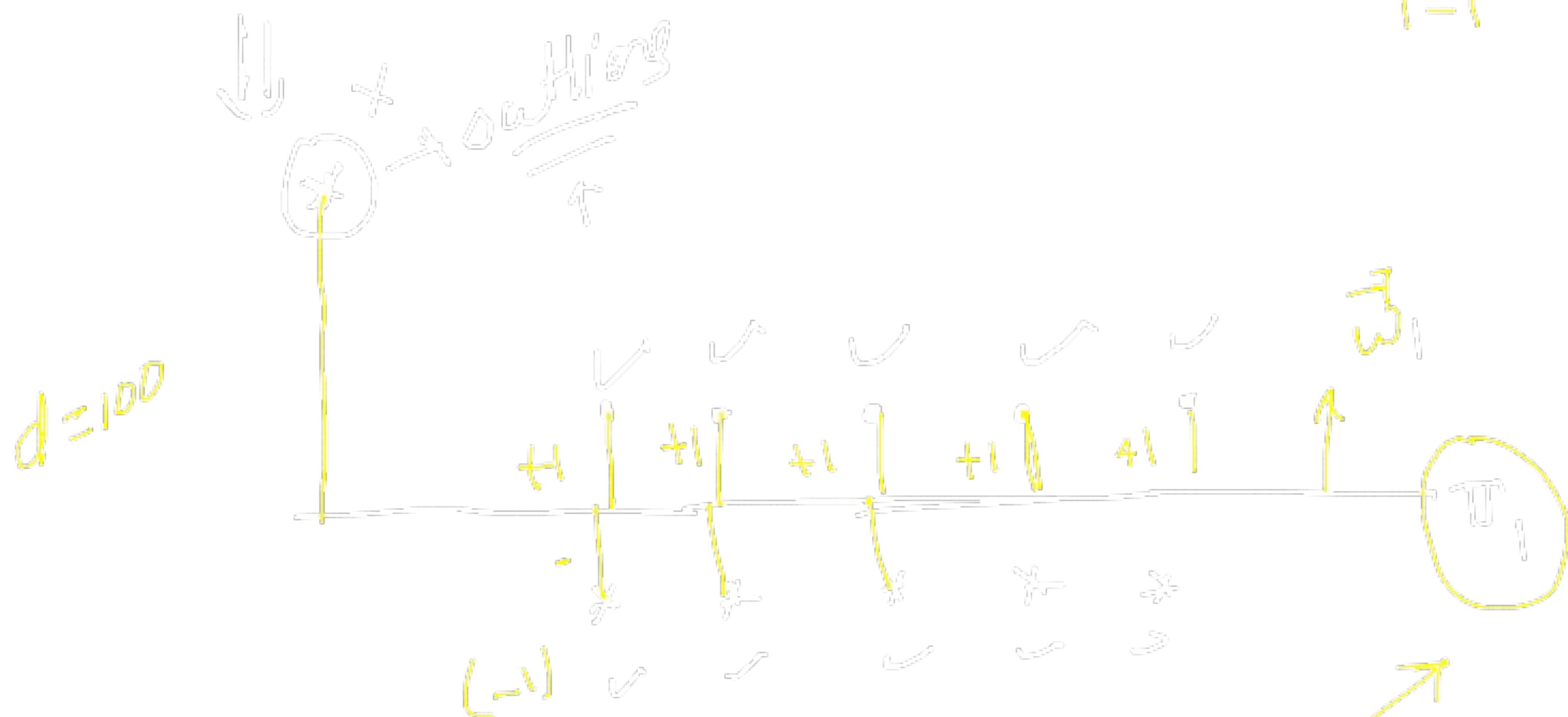
optimal

$(+w) \rightarrow \circ$

$(-w) \rightarrow *$

case(i)

$\sum_{i=1}^{n} y_i w^T x_i$



distance

$= \underline{y_i (w^T x_i)} +$

$= (+1)(+1)$

$= \overbrace{1 + (1)(1) + (1)(1) + (1)(1)}$

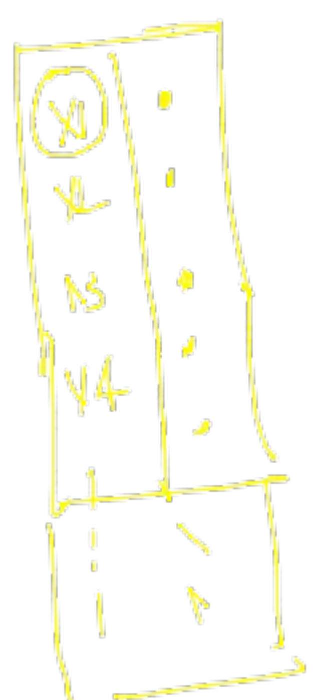
$+ \dots + (-1)(-1)$

$+ \dots$

$= 10 + (-1)(+100)$

$= 10 - 100$

$= -90$



$\frac{10}{11}$

Case ii]

$1 + (1)(2)$

$1+2+3+4+5 + (-1)(+1)$

$\Rightarrow 1+2+3+4+5-1-2-3-4-5$

$\Rightarrow 0 + (-1)(-1)$

$\Rightarrow 1$

$\vec{w}_1 \qquad \vec{w}_2$

$-90 \qquad \frac{11}{2}$

T_2

$\frac{6}{11}$

$$w^* = \arg\max \sum_{i=1}^{n} \underbrace{y_i w^T x_i}_{} + w_0$$

$\Downarrow$ $\uparrow$

f

signed distance

$\uparrow$

(large) $\rightarrow$ convinsing

$\Uparrow$

mathematical f^n

$\Downarrow$

$\Downarrow$

sigmoid f^n

$\rightarrow f(y_i w^T x_i) \rightarrow$ squeeze

small $(0-1)$

$$w^* = \underset{w}{\arg\max} \sum_{i=1}^{n} \sigma(y_i w^T x_i)$$

↑

optimal

↑

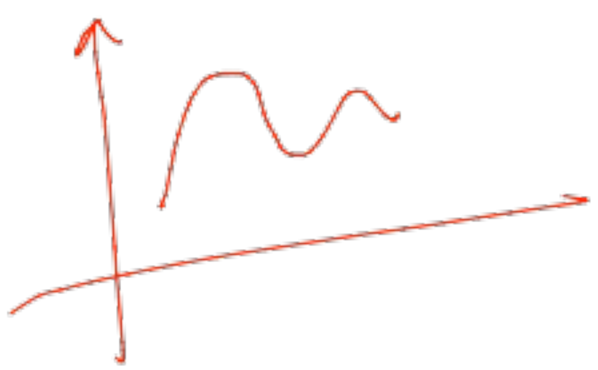
Best

$$\arg\max \sum_{i=1}^{n} \frac{1}{1+\exp(-y_i w^T x_i)}$$

↑

less impacted by

outliers



squashing $(-\infty, \infty) \Downarrow (0,1)$

Activation f^n $\boxed{NN}$

1) Sigmoid

2) tanh

3) ReLU

⋮

Sales ↑

Sales = (1) is holiday + (2) weekend + (3) discount

↑ equation

equation

constraints

sales ↑

(1) Sales → (—w) > 5000

(2) discount

w1
w2
w3
w4